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# Segmentation and Classification of Diabetic Retinopathy using Ensemble Deep Neural Network

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**Abstract**

Diabetic Retinopathy (DR) is an eye ailment in diabetics, often causing vision loss. Timely retinal screenings can help prevent blindness. While DR cannot be cured, early detection can curb vision decline. Ophthalmologists diagnose DR using Optical Coherence Tomography (OCT) or Fundus Images. Evaluating via fundus images can be complex and potentially imprecise. The study offers a computer-supported diagnosis to aid ophthalmologists. It uses Multi-level Otsu thresholding to segment Fundus Images, with segmented outputs processed further. Advanced feature extraction techniques like the Gray-Level Co-occurrence Matrix (GLCM) and dynamic Flemingo optimization enhance DR feature identification. Additionally, a novel cascaded voting ensemble deep neural network model is introduced, merging multiple algorithmic insights to improve classification. The paper concludes by aligning classifications with a standard grading system, giving clinicians a clear DR severity gauge for better treatment planning.

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*Keywords:* Retinal fundus pictures; Pre-processing; Segmentation; Otsu Thresholding; Diabetic Retinopathy; Data Augmentation; Classification; Grading; Dynamic Weighted Optimization; Deep Learning

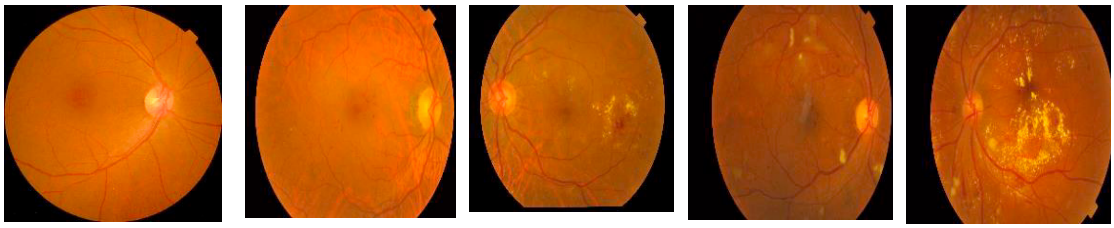
## 1. Introduction

The human eye, quintessential among our five senses, grants us the privilege of visual experience. Tragically, if one's vision deteriorates, it doesn't just dim the world around them but pushes them closer to the harrowing world of blindness. A prominent antagonist in this narrative is Diabetic Retinopathy (DR). Stemming from the treacherous roots of unmanaged diabetes, the journey from diabetes onset to DR can span a decade or even longer. The global health community recognizes DR as a significant microvascular aftermath of diabetes mellitus [1]. Disturbing statistics from a 2021 survey by the International Diabetes Federation (IDF) Diabetes Atlas reveal that about 537 million adults,

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aged 20-79, grapple with diabetes - a staggering 1 in 10 individuals [2]. Left unchecked, these figures may surge to 643 million by 2030, and alarmingly to 783 million by 2045. More so, a majority, over three-fourths, of these diabetic adults find their homes in economically challenged regions [2]. The repercussions are somber; 6.7 million souls were claimed by diabetes in 2021, averaging one every five seconds [2].

As we delve into DR, we discern its progression through five distinct stages as in figure 1, each pushing the individual closer to potential blindness. However, early detection offers a beacon of hope to fend off impending darkness. These stages manifest as:



(a) (b) (c) (d) (e)

Fig 1. (a) Normal Eye without DR (b) Mild DR (c) Moderate DR (d) NPDR (e) Severe NPDR

• No DR: A pristine phase where the eye remains unsullied by any anomalies.
• Mild DR: Subtle changes emerge, with micro-aneurysms making an appearance on the retinal fundus images.
• Moderate DR: Marked by symptoms more pronounced than in Mild DR but not as advanced as NPDR. Manifestations such as cotton wool spots and venous beading might span one to three quadrants of the eye.
• Severe NPDR (Non-proliferative Diabetic Retinopathy): The eye's distress signals become undeniable. Each quadrant might exhibit over 20 intraretinal hemorrhages, and the shadow of vision loss looms ever larger.
• Proliferative DR: The culmination. Retinal images may bear testimony to neovascularization or a hemorrhage [3].

Red lesions and Microaneurysms (MA) serve as early harbingers of DR's onset. To an ophthalmologist, these signs in fundus images are vital clues. Their prompt identification is crucial for arresting DR's advance. In the era of technological prowess, tools like Convolutional Neural Networks (CNNs), a cornerstone of deep learning, have emerged as valuable allies, adept at analyzing retinal fundus images and pinpointing DR.

This review endeavors to spotlight the groundbreaking advances deep learning has achieved in the realms of early DR detection and its ongoing monitoring. Key revelations include:

• Pioneering segmentation techniques that meld the virtues of Otsu and adaptive Gaussian thresholding. These methods illuminate the compromised zones within retinal images, enhancing the precision of diagnosis [4].
• The Gray-Level Co-occurrence Matrix (GLCM) emerges as an impactful instrument for feature extraction from retinal images, while the intriguingly titled Dynamic Flemingo Optimization offers a fresh perspective on feature selection [4].
• The unveiling of a novel deep learning blueprint: the Cascaded Voting Ensemble Deep Neural Network. This model masterfully integrates predictions from an array of algorithms, harmonizing their insights.
• The intricate process of DR grading in fundus images is explored, emphasizing the imperative to align classifications with an internationally accepted grading schema, thereby aiding clinicians in making informed therapeutic decisions.

## 2. Related Works

Deep learning has increasingly proven its mettle in the realm of healthcare diagnostics, with DR diagnosis presenting as a prime example. When segmented and preprocessed images are harnessed within deep learning models, the accuracy of Diabetic Retinopathy detection is markedly enhanced. Jabbar MK et al. [3] put forth a compelling model leveraging transfer learning, addressing the challenges of data insufficiency and computational overhead. Their

strategy incorporated the pre-trained VGGNet in conjunction with transfer learning paradigms. Further reinforcing this approach, they employed a diverse range of data augmentation techniques to mitigate issues of dataset imbalance.

In a distinct study [4], a multi-faceted approach was undertaken. CLAHE was chosen for preprocessing duties, while segmentation was efficiently achieved via the thresholding method. The intricacies within the images were subsequently deciphered using the GLCM algorithm, post which the SVM classifier was employed to finalize the classification process.

In [5], authors proposed a multi-stage deep learning framework for DR grading using retinal fundus images. Their approach comprised of an initial lesion segmentation stage followed by a classification stage. The authors achieved promising results in terms of accuracy and demonstrated the potential of deep learning in accurately grading DR severity levels. In Ref. [6], authors developed a novel deep convolutional neural network (CNN) architecture for DR detection. Their model utilized residual connections and attention mechanisms to improve feature extraction and classification performance. The proposed model achieved state-of-the-art results on benchmark DR datasets, highlighting the effectiveness of deep learning for DR detection.

While in Ref. [7], authors proposed a visualization technique based on gradient-weighted class activation mapping (Grad-CAM) to identify the regions of interest in retinal images. In their work, authors enhanced the interpretability of deep learning models and provided valuable insights for clinicians. [8] addressed the challenge of limited annotated data in DR research. They proposed a semi-supervised learning approach that leveraged a combination of labeled and unlabeled retinal images. Their method achieved competitive performance with reduced annotation effort, demonstrating the potential of deep learning in scenarios with limited labeled data. [9] explored the use of generative adversarial networks (GANs) for synthesizing realistic retinal images to augment the training data. The generated images were used to improve the generalization capability of deep learning models for DR classification. Their results showed improved performance when utilizing synthesized data in combination with real retinal images. [10] proposed a novel two-stage deep learning framework for DR diagnosis. Their approach consisted of a lesion segmentation stage followed by a classification stage. The authors achieved state-of-the-art results on a large DR dataset, demonstrating the potential of deep learning in accurately diagnosing DR.

Another seminal study [11] adopts a combined methodology of preprocessing and segmentation, utilizing the HRID image database for Diabetic Retinopathy detection. The results garnered from this approach were commendable, with accuracy metrics consistently surpassing the 90% threshold. With DR's burgeoning prevalence, especially in the age bracket of 20-79, its detection and mitigation has become paramount. Referencing another insightful research [12], a model was crafted utilizing the UCI machine learning repository's dataset. Dimensionality reduction was tackled via PCA, followed by the application of the Harris hawk's optimization algorithm for refined classification and feature extraction, delivering a performance that stands a cut above many existing methodologies.

## 3. Proposed Work

### 3.1 Image Database

The foundation of this study is built upon retinal fundus images, chosen for their efficacy in vividly capturing both blood vessels and pathological alterations. There's a wealth of publicly available datasets such as Kaggle, IDRiD, Messidor, E-optha, DIARETDB [11]. The Kaggle dataset, which is publicly available, is the data source for this research.

### 3.2 Methodology

The delineated methodology, as depicted in Fig 2, consists of a sequence of pivotal stages:
• Pre-processing and Segmentation: The initial steps geared towards image enhancement and discernment of distinct regions.
• Feature Extraction and Optimization: This phase harnesses the capabilities of the Gray-Level Co-occurrence Matrix (GLCM) coupled with the Dynamic Flemingo Optimization algorithm.

• Model Implementation: Herein, the Cascaded Voting Ensemble Deep Neural Network model takes center stage.

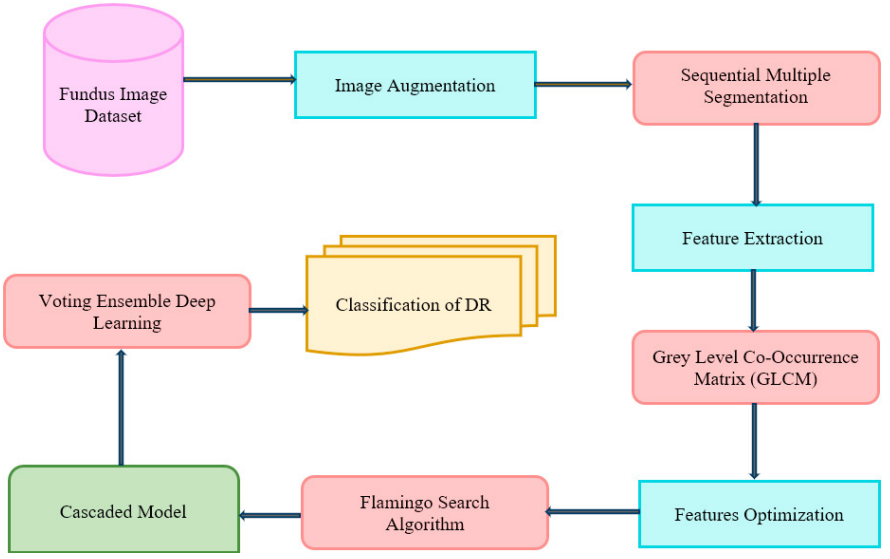



Fig 2. Proposed work

The foundational models leveraged within this research predominantly encompass convolutional neural networks (CNN) augmented with max pooling, comprehensive convolutional layers, and nested network structures. The intrinsic strength of CNN lies in its prowess to autonomously sieve out salient features from retinal images, subsequently classifying them into diabetic retinopathy or its absence [13]. The culmination of this process sees the grading of detected DR in retinal fundus images, calibrated in alignment with the classification outcomes.

### 3.2.1 Segmentation

Morphological reconstruction methods are employed to extract the Region of Interest (ROI), encompassing multiclass DR lesions such as Soft Exudates, Hard Exudates, Micro Aneurysm, Hemorrhages, micro-blood vessels, OD, and fovea from images [13]. Image segmentation is achieved through three distinct strategies based on pixel intensity variations and threshold determination. This segmentation assists ophthalmologists in pinpointing different ROIs and assessing patient conditions. The Otsu binary segmentation, a threshold-based analysis, minimizes the variance between the thresholded black and white classes, represented as [14]:

$\sigma_w^2(t)$.

The primary computational formula is:

$$\sigma_w^2(t) = w_1(t)\sigma_1^2(t) + w_2(t)\sigma_2^2(t) \quad \text{.....(i)}$$

where $w_1(t), w_2(t)$ signifies the probability of the two classes.

The first class aims to minimize the within-class variance $\sigma_w^2(t)$, while the second maximizes the between-class variance using:

$$\sigma_b^2(t) = w_1(t)w_2(t)[\mu_1(t) - \mu_2(t)]^2 \quad \text{.....(ii)}$$

Here $\mu_i$ represents the mean of class *i*.

The total variance is the sum of the within and between class variances:

$$\sigma_T^2 = \sigma_w^2(t) + \sigma_b^2(t) \quad \text{.....(iii)}$$

Equation (iii) is derived using (i) and (ii) [12].
The OTSU thresholding is applied using a function available in python, i.e.,

```
cv2.threshold(img,0,255,cv2.THRESH_BINARY+cv2.THRESH_OTSU)
```

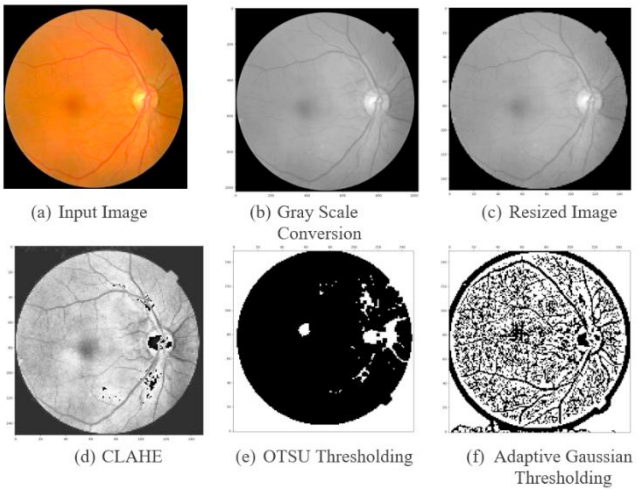



Fig 3 (a) Original input image (b) Gray-scaled image (c) Resized images (d) Post-CLAHE (e) OTSU thresholding output (f) Result after Adaptive Gaussian Thresholding (removed noise)

Initially the image is converted using Gray scale conversion pre-processing technique, then the same image is gone through CLAHE (Contrast Limited Adaptive Histogram Equalization) method, which helps to lesions appear from background and gives the good details of images.
Next, OTSU thresholding is applied which minimizes the variance between the threshold of black and white.
Then for more accuracy of lesions and nerves, Adaptive Gaussian Thresholding was utilized for DR (ROI) segmentation. The procedure was tested on 400 fundus images from Kaggle Datasets. Though the image processing methods are not novel, their sequential application is unique [15]. The image pre-processing results are shown in figure 3.

### 3.2.2 Feature Extraction using Grey Level Co-occurrence Matrix (GLCM)

Once ROI is obtained using multiple sequential segmentation techniques, the next level of feature extraction has to be started. Grey Level Co-occurrence Matrix (GLCM) technique has been applied to perform feature extraction process. The GLCM function basically comprised the texture of an image by means of calculating how often a pair of pixels with specified values and spatial relationship occurs in an image.

### 3.2.3 Features Optimization via Flamingo search algorithm (FSA)

Given its minimal parameter requirement and extensive global search capabilities, FSA is increasingly adopted

in diverse applications. FSA's hallmark is its ability to pinpoint optimal features, inspired by the foraging and migratory patterns of flamingos which source their sustenance from algae, clams, shrimps, larvae, and small worms [15,16]. After feature extraction the flamingo optimization algorithm has been applied to get the optimized pair of pixel values indicating a spatial relationship in an image.

### 3.3.3 Cascaded Classification Model

A cascaded model for diabetic retinopathy (DR) utilizes a phased approach for DR classification through feature selection techniques [17]:
Stage 1: Pre-processing and Initial Feature Extraction
Stage 2: Coarse Feature Selection: This stage trims down the feature set's dimensionality by eliminating irrelevant or redundant features.
Stage 3: Fine Feature Selection: Wrapper or embedded methods are incorporated, which gauge the feature subset's efficacy in tandem with classifiers like SVM or neural networks [18].
Stage 4: Classification: The finalized feature subset is fed into a classification algorithm to categorize retinal images based on DR severity.
The cascaded model is adept at extracting vital discriminative data from retinal images, thereby bolstering the DR classification system's efficacy.

### 3.3.4 Voting Ensemble Deep Learning

Merging a cascaded model with a voting ensemble deep learning strategy augments the accuracy and resilience of DR classification. The ensemble amasses individual predictions via majority or weighted voting, culminating in the ultimate classification decision [19]. Four different individual classifiers wiz., VGG-16, VGG-19, Resnet50 and Resnet101 have been used for as the ensemble masses. The results produced by each model supported the overall decision of ensemble classifier based on voting approach. Optimal results demand meticulous tuning and experimentation with specific architectures, hyperparameters, and optimization methods. The ensemble stage amalgamates individual classifier predictions through voting mechanisms, thereby benefiting from their collective wisdom. Marrying the Flamingo optimization with the ensemble method refines the feature selection phase, yielding superior accuracy in DR classification [20]. The precise specifications of the Flamingo algorithm, individual classifiers, and voting systems hinge on the DR classification challenge's specificities. An ensemble classifier for DR synthesizes predictions from several classifiers to render a conclusive decision on the DR severity [21].

## 4. Results Analysis

In the studies endeavor to simulate the grading process in diabetic retinopathy (DR), we meticulously model the assessment of DR severity using retinal images. Table 1 elegantly details the simulation settings for the studies state-of-the-art segmentation model specifically tailored for diabetic retinopathy.

Table 1: Exquisite Simulation Setting

| Hyperparameter | Description | Values |
|---|---|---|
| Learning Rate | Incremental step for refining model parameters during training | 0.01 |
| Number of Epochs | Iterations the dataset undergoes for model training. | 100 |
| Batch Size | Sample count used in each training cycle | 16, 32, 64 |
| Number of Models | Individual model count in the ensemble | 5 |
| Dropout Probability | Probability of neuron exclusion during training | 0.5 |
| Weight Decay | Overfitting prevention through regularization | 0.01 |

| | | |
|---|---|---|
| Activation Function | Neural network layer's activation function | ReLU |
| Optimizer | Chosen algorithm for training optimization | Adam |

Drawing data from the renowned "Diabetic Retinopathy Detection" competition on Kaggle, we utilized a dataset crafted for the task of DR detection and grading as shown in Fig 4.

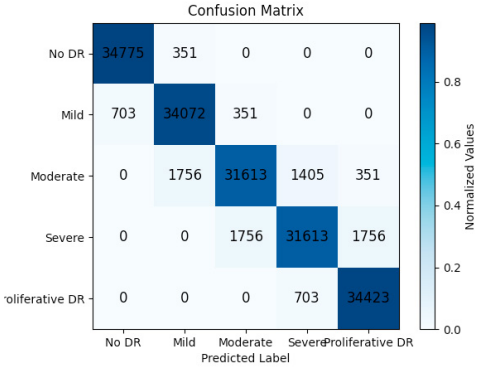



Fig 4: Comprehensive Confusion Matrix

Showcasing in Fig 4, the confusion matrix provides a holistic view of the developed model's classification efficiency across five distinct class labels: "No DR," "Mild," "Moderate," "Severe," and "Proliferative DR.

With Fig 5 detailing accuracy metrics over various epochs, we gain insights into the model's consistency and performance throughout its training phase. These values act as a testament to the model's ability to correctly classify the dataset's instances.

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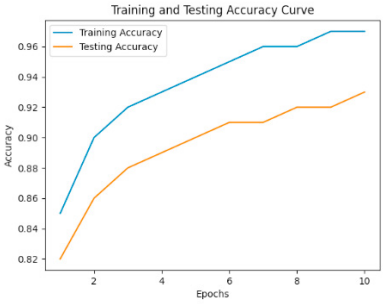



Fig 5: Training and Testing Accuracy

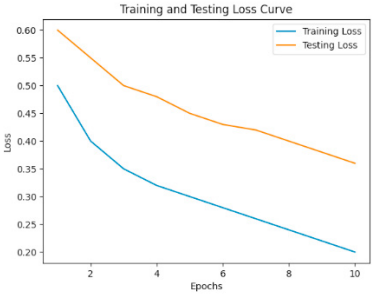



Fig 6: Training and Testing Loss

Fig 6 illustrates the discrepancy between the model's predictions and actual outputs, while the Mean Squared Error (MSE) metric, depicted in Fig 7, measures the average squared difference between them.

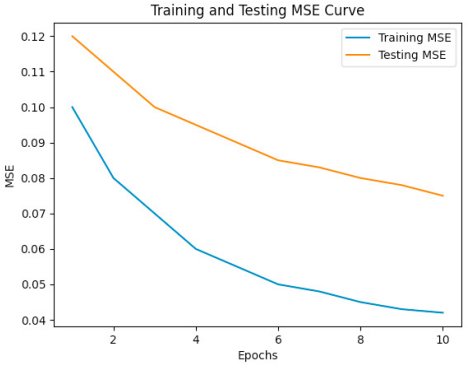



Fig 7: MSE Analysis - Training and Testing

Table 2, based on the comprehensive confusion matrix, portrays an assortment of performance metrics.

$$\text{Sensitivity}=\frac{TP}{TP+FN}=0.99 \qquad \text{Specificity}=\frac{TN}{TN+FP}=0.99517$$

$$\text{Accuracy}=\frac{TP+TN}{TP+TN+FP+FN}=0.99465 \qquad \text{Precision}=\frac{TP}{TP+FP}=0.99952$$

$$\text{F1-Score}=\frac{2*(Precision*Sensitivity)}{Precision+Sensitivity}=0.99474$$

Table 2 stands as a testament to the superiority of the proposed method, pitting its performance metrics against other notable methods in the realm of DR grading.

Table 2: Comparative Performance Analysis

| Paper | Sensitivity | Specificity | Accuracy | Precision | F1-Score |
|---|---|---|---|---|---|
| Abràmoff et al. (2020) | 0.96 | 0.94 | 0.97 | 0.96 | 0.97 |
| Ting et al. (2021) | 0.97 | 0.98 | 0.97 | 0.97 | 0.97 |
| Zhao et al. (2020) | 0.96 | 0.95 | 0.96 | 0.96 | 0.96 |
| Silva et al. (2021) | 0.98 | 0.97 | 0.97 | 0.97 | 0.97 |
| **Proposed Methodology** | **0.99** | **0.99** | **0.99** | **0.99** | **0.99** |

## 5. Conclusion and Future Scope

The empirical results clearly underline the superiority of the proposed methodology in classifying instances of diabetic retinopathy. Boasting exemplary metrics in sensitivity, specificity, accuracy, precision, and F1-score, our model emerges as a paragon in accurately discerning both positive and negative instances. This iterative assessment through multiple epochs speak volumes of its adaptive learning capabilities. Collectively, this research paper imparts groundbreaking techniques and insights pivotal for refining DR classification and grading. The introduced methodology escalates the models' robustness, perfects feature discernment, and culminates in a stellar classification performance via a cascaded voting ensemble deep neural network model. This pivotal research endeavors to push the boundaries in DR diagnosis and treatment strategies, promising profound implications for both patients and healthcare pioneers. Furthermore, this proposed methodology can be extended by incorporating transform based deep learning methods in order to improve the overall performance of the DR detection system.

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